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Review

An inclusive approach for organic waste treatment and valorisation using Black Soldier Fly larvae: A review

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ABSTRACT

Solid waste management and its stabilization are a sophisticated task and the most challenging one as it requires improved collection and treatment strategies. From the past decades, there is a huge emphasis on valorisation of waste along with its management i.e. an integrated and income generating sustainable approach for solid waste management. Use of Black Soldier Fly (Hermetia illucens) in organic waste composting is a novel and an environment friendly approach which holds enormous potential and therefore, is strongly captivating people's attention worldwide. The Black Soldier Fly (BSF) larvae composting is a self-sustained cost-effective method promoting high resource recovery and generating value added products thereby developing new economical niches for industrial sector and entrepreneurs in developing countries. Here, we reviewed the importance of BSF larvae in organic waste treatment and delineated the life cycle patterns, feeding habits and environmental conditions affecting the survival of the species. This review paper has also congregated the efficiency of BSF larvae to compost different types of organic wastes or biomasses and a portion of various possible end applications of these avid eaters. Through this literature review the authors have also made an attempt to evaluate the present constraints, research gaps and future directions associated with this technology. BSF larvae composting is a comprehensive approach indeed providing the waste an aforementioned value wherein technological innovations can boost up the efficiency of system. Thus, the present study is an aggregate of applications of BSF larvae for societal benefit in a holistic way.

1. Introduction

Solid waste sector, owing to rapid urbanisation, demographic changes and consumer behaviour and its subsequent negative social, economic and environmental impacts, has not only captivated but also confronted the municipalities and decision makers including the concerned mass with new challenges towards its sustainable and economically viable management (Diener, 2010). According to United Nations report, the current population of world is about 7.6 billion and is expected to reach 8.6 billion in 2030 and 9.8 billion in 2050 whereof half of the population increase is expected to be in countries like India, Pakistan, USA, Nigeria, Uganda, Indonesia, the Democratic Republic of the Congo, the United Republic of Tanzania and Ethiopia (World Population Prospects,). Population growth with such high rate demands better standard of living, consequently contributing to solid waste generation in huge quantities either directly or indirectly. Urban sprawl, economic growth, collection and treatment efficiency of a system acts as the governing factors that greatly influence the quantity and complexity of the waste generated. Global waste generation is expected to increase from 2 billion tons in 2016 to 3.4 billion tons in 2050 (Kaza et al., 2018) with Asian and African countries as the major contributors. Approximately 1.3 million tons of solid waste is generated per day by world cities and is expected to increase up to 2.2 billion tons by 2025 (Hoornweg and Bhada-Tata, 2012), whereof Asian urban cities generate about 760,000 tons of municipal waste per day with projected increase by multiple folds (about 1.8 million tons' waste/day) by the year 2025 (UN-Habitat, 2010).

The major lacunae in waste management lies in inefficiency of present technologies to promote valuable utilization of waste in reuse and recycling system. The current management policies of solid or municipal waste is greatly restricted to collection, transportation, treatment and disposal and therefore, lacks large scale valorisation of organic rich waste (municipal solid waste). In this perspective, researchers across the globe are incessantly trying to develop and find

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sustainable ways of waste decomposition and determine its inherent nutrient and economic value.

Inadequate management of organic waste are the major problems in developing countries that may result in devastating environmental and anthropogenic impacts (Hoornweg and Bhada-Tata, 2012). For instances, in many countries landfill operations are still the common way to manage waste due to its economic advantages but equally contribute to environmental hazards such as generation of heavily polluting leachates, land subsidence, epidemiological hazards, etc. (Renou et al., 2008). On the other hand, waste incineration also has potential health impacts. The workers associated with incineration are more often at risks of different cancer and other hazards (Hu and Shy, 2001). The release of toxic gases from incinerators are other concerns related to its operations. Thus, solid waste management system needs to be well equipped and systematized and countries shall look for economical and technically feasible strategies. The organic waste forms the maximum portion of the solid waste and thus have high nutrient values which may serve for the welfare of society. Capturing this inherent value of organic waste could alleviate the environmental consequences to greater extent and simultaneously serve for the purpose of food and fodder or other products of commercial interest. (Lalander et al., 2019). Moreover, organically rich biomass has great potentials to replace chemicals and can be used as fertilizers for agricultural crops (Diacono and Montemurro, 2011). The present technologies of waste decomposition and management are lagging behind in one or the other way (such as incineration or land filling) and responsible for various environmental impacts. Confronted with such problems, the application of BSF, *Hermetia illucens* is emerging out to be very efficient green technology in solid waste management. The feeding activity of the species has the capability to remarkably reduce huge volume of organic biomass and concurrently can offer valuable animal or human feed having high nutrient composition (Diener, 2010).

In another aspect, to cope up with elevating global human population and sustain such huge population, demand and supply need to be balanced, and in this context, food production is estimated to increase by 70–90% of the current pace (Elferink and Schierhorn, 2016; Lalander et al., 2019). Moreover, the consumption trends are highly varied for this generation wherein the demand for livestock, horticulture, fisheries poultry meat and relevant products is markedly high over cereals. As a result, massive decline is witnessed in cereals demands over the last two decades (Edmeades et al., 2010; Van Huis et al., 2013; Kumar, 2016). According to a report, in a comparative analysis annual cereals production is projected to increase by 3 million tons by 2050 from 2.1 million tons whereas the meat production from 200 million tons to 470 million tons which is an extremely big figure (FAO, 2009). Increased livestock products consumption is another issue which requires high protein animal feed in return (Lalander et al., 2016). The BSF larvae have high protein value and high nutritive composition and therefore regarded as an excellent source of animal feed for livestock raising. Thus, BSF larvae may be exploited or preferred over other conventional feed to promote livestock raising. However, the considerable progress has been achieved in animal feed production, the present conventional technologies are still not economically viable in many aspects. The requirements of huge land area and high technical skills are the major issues associated with the technologies. The consumption of major proportion of global human water use (about 8%) is another challenging issue threatening the sustainability and economic viability (Amata, 2014). Subsequently, high price of conventional animal feed resources, poor quality feed and the major gap between the demand and supply are other constraints. To meet the requirements, non-conventional feed resources are being developed from agricultural crop residues and agro-industrial by-products but processing of such resources is a difficult task indeed (Amata, 2014). Creating additional value chains, BSF larvae composting is gaining huge attention as a sustainable alternative for animal husbandry, livestock raising, feed for aquaculture and thereby strengthening economic resilience of small-scale industries (Diener, 2010; Cickova et al., 2015). These flies can be reared on waste, promoting the recycling of waste and reducing its environmental footprint just like any other insect species (Smetana et al., 2016). BSF larvae can feed on a variety of waste including the kitchen waste (Diener et al., 2011a; Nguyen et al., 2015), poultry waste (Zhou et al., 2013), dairy manure (Myers et al., 2014) and human faeces (Lalander et al., 2013). Thus, BSF larvae can alleviate the problems of solid waste management concurrently supplementing the livestock products or feed with high nutritive assay.

The species also have the potential candidature for production of other supplements such as biofuel production. Henceforth, BSF larvae technology contributes to a self-sustained cost-effective treatment method. However, the economic feasibility of waste management depends on many factors such as type of organic source, nutrient availability, waste to biomass conversion ratio, etc. (Lalander et al., 2019). But indeed, organic waste conversion by BSF larvae is an outstanding recycling technology and a comprehensive approach as it has lower environmental impact (less greenhouse gas emissions) and low ecological footprint for production of protein feed or other nutrient supplements. Thus, this technology is suitable and can be used extensively in low- and middle-income countries that need low capital investments (Van Huis et al., 2013). From this aspect, the present study highlights the importance of BSF larvae for sustainable organic waste management and in production of animal feed and other value-added products in a wholesome way.

2. Methodology

This review paper is entirely based on secondary data available in scientific domain focused on the respective sub-study themes. The authors have tried to compile the literature available on the life history traits of the species BSF, its crucial functioning in organic waste treatment published in peer-reviewed journals having wider reach to researchers, books, periodicals, abstracts, indexes, research reports, conference papers, and other technical reports. The authors searched a wide range of keywords in online database tools and scientific domains of Science Direct, Research Gate, Google scholar, Web of Science and Scopus (up to January 2019) to establish versatililty of *Hermetia illucens,* (BSF) in waste composting (Bramer et al., 2017). This paper is a result of 133 research publications which met the inclusion criteria for the review paper. Published data were listed in tables and used as a tool to discuss the reported potential of BSF larvae, the fervent gourmand of organic waste and their significance as value chain developers. In brief, this review is an effort to strengthen support for existing theories and define or suggest new research directions, based on the previous studies available in the area of BSF larvae compositing and valorisation.

3. The Black Soldier Fly species (*Hermetia illucens*)

3.1. Geographical distribution of the species

Black Soldier fly (*Hermetia illucens*) are among one of the 5 genera belonging to sub family Hermetiinae under order Diptera (Woodley, 2001) (Table 1). The other four genera are namely *Chaetosargus, Patagiomyia, Chaetohermetia* and *Notohermetia*; *Hermetia illucens* is the most extensively distributed species among all. The species has the cosmopolitan distribution in tropical and warm temperate regions (between about 45°N and 40°S) from Neotropical to Australasian, Nearctic, Palaearctic and Afro-tropical Regions (Fig. 1) (McCallan, 1974; Ustuner et al., 2003). However, the species was very first reported in Hawaiian Islands in a Hilo Sugar company and is native to America (McCallan, 1974; Gujarathi and Pejaver, 2013).

3.2. Basic anatomy and the life cycle

These flies are large slender black coloured species with brownish

Table 1
Taxonomical classification of Black Soldier fly (*Hermetia illucens*).

Taxonomical Classification

Kingdom	Animalia
Phylum	Arthropoda
Class	Insecta
Order	Diptera
Suborder	Brachycera
Superfamily	Stratiomyoidea
Family	Stratiomyidae
Subfamily	Hermetiinae
Genus	Hermetia
Species	*H. illucens*

tinged wings and antenna (three segments) projecting outwards from head. The antenna consists of 8 flagellomeres and the last flagellomere is as long as the antenna (Ustuner et al., 2003). The adult body is divided into head, thorax and abdomen like other dipterian species. The eyes are bare and broadly separated. The abdomen consists of 5 segments with white abdominal spots. Females are generally shorter than males but have large terminalia than male genitalia and also larger wings than males. The length of the female body and wings generally varies from 12 to 20 mm and 8–14.8 mm (Ustuner et al., 2003).

The life cycle basically consists of five stages: eggs, larvae, pupae, prepupae and adults. The larval and pupal stages contribute maximally to the entire life cycle whereas adult and eggs hatching stages are relatively short (Fig. 2). As the life span of BSF is very small; a single female yields a large number of eggs which soon hatch into neonates' larvae. These flies have very short life span of about 6–8 days which may be extended to 6–7 weeks depending upon the environmental conditions, as these flies slow down their activity under unfavourable conditions (Tomberlin and Sheppard, 2002; Alvarez, 2012; Caruso et al., 2013; Dortmans et al., 2017; Banks, 2014). They may attain flexible life cycle by extending their larval stage (reportedly up to 4 months) under low temperature and low food availability (Furman et al., 1959). This facilitates the viability of larval population during the shortage of organic waste (source of food for neonates). Nevertheless, egg production and the development of larvae largely depend on the quality of food (Chippindale et al., 1993).

The adult females mate only once in life time and therefore single oviposition event occurs throughout the lifetime. Mating occurs after two days of emerging, and consequently oviposition occurs in dry crevices near the food source. They lay eggs two days after the copulation which soon hatch into neonate larvae. These larvae migrate to their available feeding source. As, most of the feeding occurs at this stage, the larvae soon transform into last immature instar i.e. prepupae and feeding stops. Prepupae migrate away from the food source to the dry crevices, for pupation. At this stage, the species attain its maximum size consisting of 36–48% protein and 33% fat (St-Hilaire et al., 2007a; Yu et al., 2009) and finally metamorphosis completes when adults develop after 14 days (Diener, 2010; Pathak et al., 2015). The adult flies do not possess mouth parts or digestive system or the stinger and therefore do not pose threat to living beings (Park, 2016).

3.3. Feeding habits

The species feed only during the larval stage, and they feed voraciously on decaying plant and animal material such as decaying fruits, vegetables, rotting food products, heaps of cow, poultry manure, human and animal waste (Ustuner et al., 2003). These larvae feed on organic waste double the weight of their own and because of this inherent quality among the BSF are popularised as a sustainable means of reduction of organic waste by higher volume. The adults don't feed and are found to be established over tree trunks, walls, windows and garden

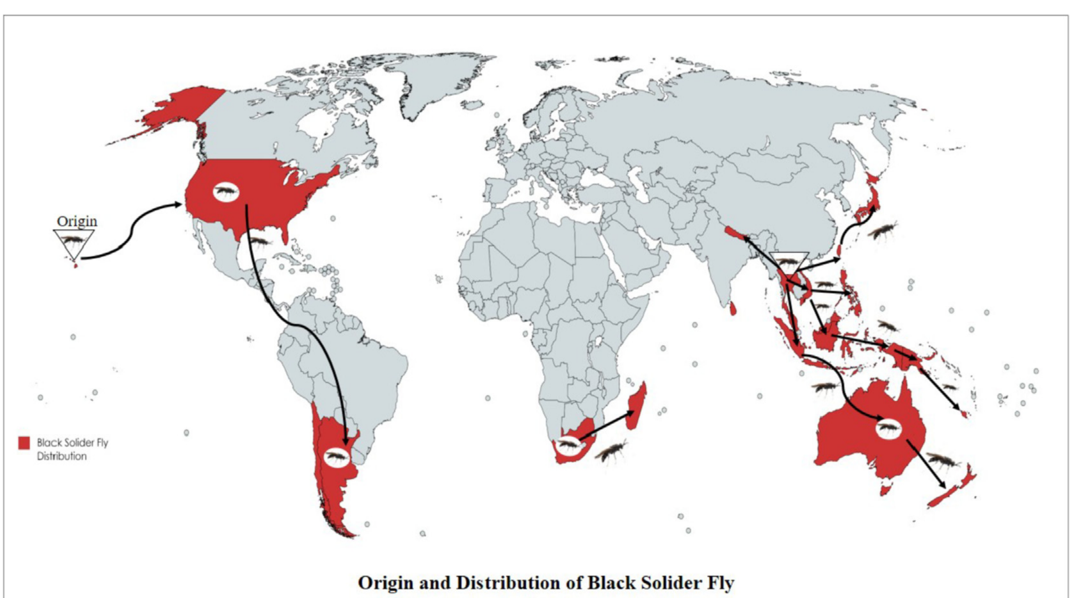

Fig. 1. Geographical distribution of Black Soldier Fly

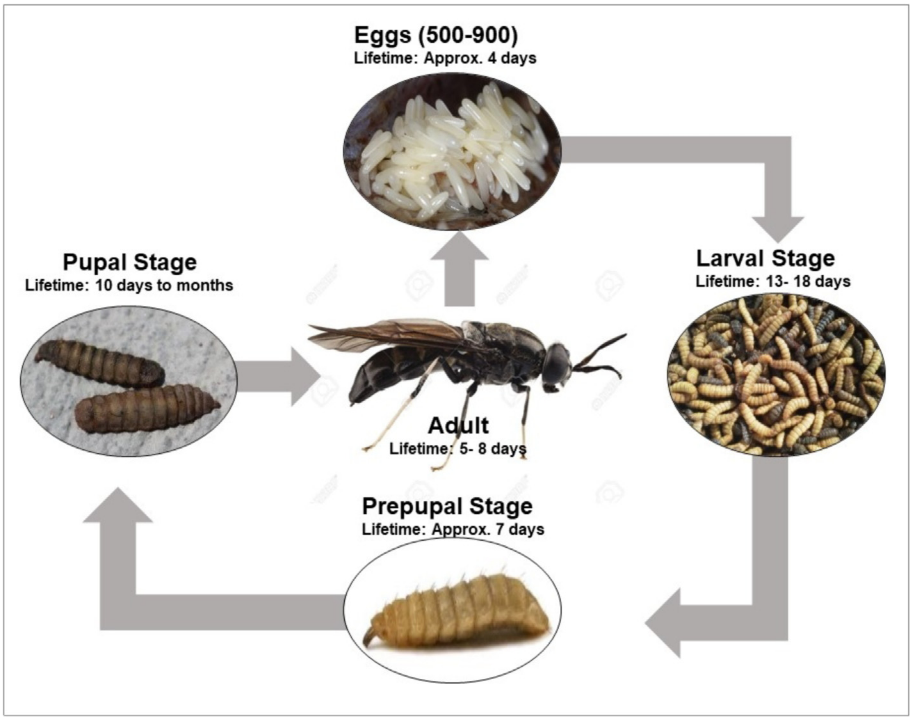

Fig. 2. Typical lifecycle stages of Black Soldier Fly

plants in residential areas (Rozkosny, 1983; Ustuner et al., 2003).

Wastes with high organic matter serve to be the absolute source of food for BSF larvae. This is the reason the adult flies lay their eggs in dry crevices near the moist organic waste or dumpsite or compost site (serving to be the breeding site). It is during the larva stage when they feed as much as they can, because once they pupate and transform into adults, they stop feeding and thereafter rely solely on the fat reserve of the body. Interestingly, these species do not have affinity towards human habitation or food; therefore, do not act as a vector for spread of diseases (Bradley and Sheppard, 1984; Schremmer, 1986; Leclercq, 1997; Sheppard et al., 2002).

4. Factors affecting the life cycle of Black Soldier Fly (*Hermetia illucens*)

The abiotic factors such as temperature, relative humidity, light source and the other waste related attributes (such as moisture level and pH) greatly influence the oviposition and development of larvae. The presence of optimum conditions enhances the treatment efficiency of the of BSF larvae and also the species growth and development.

4.1. Temperature and humidity

BSFs are eurythermal species which can tolerate wide range of temperatures (15–47 °C) but at the same time are extremely sensitive (Park, 2016). Temperatures and relative humidity greatly influence the development, mating and oviposition of the species (Barry, 2004; Tomberlin et al., 2009). Studies conducted by various group of authors showed that 99.6% of the oviposition occurred in the temperature range of 27.5–37.5 °C and 60% relative humidity conditions (Booth and Sheppard, 1984; Sheppard et al., 2002; Holmes, 2010). In the same context, another group of authors reported that 50–90% relative humidity is optimum for development of BSF in laboratory conditions (Bradley and Sheppard, 1984; Diener et al., 2009). Similarly, Tomberlin et al. (2009) in his study found that at a temperature of 27 °C, the development was most efficient for both males and females at all stages but at higher temperatures (30–36 °C) metabolism increases and adults remain small and consequently smaller lifespan, where the development is largely restricted or inhibited. He also found that at 27 °C the reared flies were about 5% more in weight and 10% longer that the flies reared at 30 °C. It was further investigated by many authors that temperature of 27 °C and ambient relative humidity (about 60%) maintained at the site are the optimum ideal conditions for mating and egg laying (Sheppard et al., 2002; Holmes, 2010). The findings of the study were also positively correlated with one another.

Moreover, it was also noted that temperature and humidity can have serious consequences on egg eclosion and colony development if not maintained optimally (Park, 2016). In the same perspective, Holmes et al. (2012) also earlier concluded that 25% relative humidity can have higher desiccation rates and high mortality of species but species subjected to 70% relative humidity have 2–3 days' longer life span.

4.2. Moisture content

The growth and survival of the species is highly influenced by the moisture content of the waste (Cheng et al., 2017). As reported by many authors, the presence of excessive moisture content can hinder the decomposition rate and the resulting residue may be accompanied with thick and clumpy material causing difficulty in further processing (Diener et al., 2011b). Proper moisture control helps overcoming such problems. Fatchurochim et al. (1989) in his study concluded that the survival efficiency of filth fly fed with poultry manure was highest in the presence of moisture content between 40 and 60%. Similarly, highest growth of pupae was recorded at 85% moisture present in the faecal sludge in the study of Banks (2014). Another group of authors also concluded that 80% moisture content in food waste is optimum for BSF growth (Cheng et al., 2017; Dortmans et al., 2017).

4.3. pH

pH is an intrinsic parameter affecting the lifecycle and survival of BSF. Numerous studies on dipterian flies have revealed that, pH above 6 is optimum for larval development and growth (Meneguz et al., 2018) however, the study on effect of pH particularly on BSF larvae is limited to the best of our knowledge. Green and Popa (2012) investigated the effect of organic leachate and found that BSF larvae were able to regulate the pH up to 9 of the liquid organic leachates. However, it was also concluded by Alattar (2012) that the ability to regulate the pH of organic medium is strictly dependent on the larval density. Ma et al. (2018) has further investigated the impact of different pH levels on larval development and pointed out that the pH above 6 and up to 10 is most appropriate for larval growth performance and larvae have higher weights as compared to larvae subjected to pH level 4 or 2. The author also concluded that the larvae were able to regulate the pH of alkaline substrate from 8 to 8.5 but not of highly acidic substrate.

4.4. Light source

For mating events in BSF, direct sunlight plays major role in the natural environment and this is the reason that significant mating does not occur in winter seasons. The indoor experiments require artificial lightning source. About 85% mating events occur in the presence of natural sunlight with intensity of 110 μmol m^2s^{-1} (Park, 2016). In a study, artificial light source is reported to have influence on reproductive events of the species. This particular method can be very advantageous and effective for rearing the species outside their native habitats, where sunlight is the main influencing source. The mating and oviposition of the species was positively correlated with the use of Quartz-iodide lamp (135 μmol m^2s^{-1}) as the artificial light source (Zhang et al., 2010). Insects do not see the light past 700 nm, so, a wavelength range between 450 and 700 nm is optimum for reproductive events of adults (Briscoe and Chittka, 2001; Zhang et al., 2010). However, to have better understanding of biology of species under the influence of light, additional research is required for identification of appropriate spectrum.

5. Black Soldier Fly larvae as biological means of organic waste treatment

CORS (Conversion of Organic Refuse by Saprophages) technologies for organic waste treatment offer feasible and economically viable means of treatment. It involves technologies such as vermicomposting, bacterial decomposition and decomposition by biological species (Table 2). Not only that, such sustainable technologies offer promising alternatives of nutrient recovery from waste (Elissen, 2007). BSF larvae treatment technology is one such technology that integrates three major terms waste composting, nutrient recovery and income generation in its application (Sheppard et al., 1994). It is far better alternative in comparison to other composting techniques considering the low cost, low maintenance, less sophisticated and easy operation, land requirements, low ecological footprints and more adapted economical potential.

BSF larvae exhibit a number of social, economic and environmental benefits to the society, among which massive organic waste reduction is the foremost and sustainable means of solid waste management. Diener et al. (2011a) in his findings reported 65–78% of total waste reduction depending upon the amount of waste added daily. In Indian climatic conditions, feeding on the kitchen waste, Pathak et al. (2015) also reported that larvae were able to reduce the waste by 50% of original volume. Besides that, the larvae are also regarded as rich source of proteins and fats and are consumed as food for animal feed and to some extent by humans. The high lipid content of larval stage is also reported to have application in biofuel production. The potential benefits of BSF larvae are represented below in diagrammatic form (Fig. 3).

As the larvae feed voraciously on organic rich waste, it results in highly reduced weight and volume of waste in a very short duration of time. The BSF larvae technology is economically feasible since the species are self-harvesting and require less technical skills. The major findings of different authors also revealed that the species play vital role in modifying the microflora present in the waste while composting and also reduces harmful bacteria such as *E. coli* and *Salmonella enterica* by secreting harmful bactericidal compounds (Erickson et al., 2004; Gabler and Vinneras, 2014; Lalander et al., 2015). The flies are also reported to naturally control the population of houseflies and blow flies by repelling their oviposition (Furman et al., 1959; Bradley and Sheppard, 1984; Schremmer, 1986; Leclercq, 1997). Another compelling feature of this fly is that adults do not feed and therefore are non-pathogenic (Sheppard et al., 2002). Not only this, despite heavy metal contamination and adverse conditions and subsequent decreased fecundity, the BSF larvae have enormous potentials for organic waste composting and management. In a study, heavy metal accumulation (cadmium, lead and zinc) did not had any significant effect on the lifecycle of BSF (Diener et al., 2015). The findings were also in positive correlation with other researchers (Ortel, 1995; Maryanski et al., 2002). BSF larvae has also showed suppressed spread and distribution of medicinal drugs such as carbamazepine, roxithromycin and trimethoprim and pesticides (azoxystrobin and propiconazole) into the environment with no evidences of bioaccumulation in the larvae (Lalander et al., 2016). Recently, in 2018 the larvae are also reported to have the capability to degrade 97% of tetracycline, wherein the intestinal microflora (Bacteroidetes, Firmicutes and Proteobacteria, Basidiomycota, Lysurus, Trichophyton and Entyloma) played vital role in degradation process (Cai et al., 2018). But the fact that the chemicals or heavy metals can greatly impact the life history traits of the species and eggs development cannot be ignored.

The larvae are capable of feeding on variety of organic waste. They have been reported to feed on rice straw, distiller's grain, animal offal including food waste, faecal sludge and kitchen waste. Diener et al. (2009) stated that the BSF larvae are highly effective in reducing the huge volume of organic waste and converting the waste to protein rich biomass at the same time which can greatly contribute towards sustainable agriculture. A study conducted at NC State University (2006) showed that the BSF larvae feeding on pig manure not only reduced the biomass but also the nutrient content of the pig manure. About 85% nitrogen and 75% phosphorus was reduced from the manure. In another study by a group of authors, cow manure showed reduced nitrogen and phosphorus content by 30–50% and 61–70% (Myers et al., 2014). While in a study of Costa Rica, BSF larvae were capable of reducing dry matter by 68% and despite the presence of zinc contaminated feed and hostile environmental conditions; it was possible to harvest an average of 252 prepupae per day (Diener et al., 2011a). However, the contamination has affected the fecundity rate of the species to some extent (weak in appearance and susceptible to additional stress factors). Some important findings during the studies also revealed that, the presence of proper drainage system and a well-balanced feeding under proper environmental and nutritional requirements can boost the efficiency of BSF larvae composting system. The larvae are also claimed to significantly reduce the sludge biomass when subjected to pure faecal sludge (Diener et al., 2011a). But the treatment efficiency of the BSF larvae is highly influenced by substrate properties as it greatly affects the larval development and waste to biomass conversion ratio. In the same context, Lalander et al. (2019) concluded that BSF larvae are adaptable to variety of organic waste such as Abattoir waste, human faeces, food waste, and more preferably the mixture of fruits, vegetables and abattoir waste (Lalander et al., 2019). BSF larvae are being employed as a composting agent by many authors for treatment of pig faeces, coffee pulp and other types of faecal sludge from past many decades (Larde, 1990; Newton et al., 2005a). Newton et al. (2005a) in his study achieved pig faeces fresh material reduction by 56%. Bradley (1930) found the BSFs breeding in privy pits of Louisiana, USA.

Pathak et al. (2015) reported the efficiency of BSF larvae to decompose kitchen waste in Indian climatic conditions with optimum requirements. However, the compost obtained after decomposition was low in nutrient. They also reported that it is not possible to do vermincomposting in presence of larvae and this is one of the important

Table 2
Biological Composting methods in practice.

S. No.	Methods	Key factors	Advantages	Disadvantages	References
1.	Aerobic composting	Microbes, aerobic conditions, moisture, temperature, pH and nutrients porosity	1 Highly efficient technology 2 Reduced odour problems under optimum aerobic conditions 3 Heavy metals reduction	1 Total nitrogen and phosphorus losses 2 Generation of carbon dioxide and methane under low aeration condition	Parkinson et al. (2004); Liu et al. (2007); Bernal et al. (2009); Kumar et al. (2018)
2.	Windrow system		1 Low cost methodology 2 40–50% waste reduction efficiency 3 Methane reduction by 64%	1 Slower rates of decomposition without mechanical agitation 2 Highly labour-intensive technology 3 Moisture and temperature control are difficult 4 Large land space requirements	Gajalakshmi and Abbasi (2008); Cayuela et al. (2008); Fountoulakis et al. (2009); Manyapu et al. (2017); Kumar et al. (2018)
3.	Aerated static pile composting		1 No turning of waste required 2 Mechanical operation, less labour-intensive technology 3 Requires low management and fewer equipment 4 Cheaper in terms of manpower and cost	1 Need good structure to maintain good porosity in the waste pile, avoid poor air distribution and uneven composting 2 Long process and requires large land area	Misra and Roy (2007); Gajalakshmi and Abbasi (2008); Manyapu et al. (2017);
4.	In vessel composting		1 Advanced technology equipped with exhaust air control system, mechanical turning devices and aeration pumps 2 Better process efficiency, optimization and process control 3 Small space requirements 4 Thermophilic condition can be achieved	1 Expensive than other conventional systems due to machinery and equipments	
5.	Vermi composting	Earthworms, microbe's aeration, temperature, moisture, pH and nutrient availability	1 Ability to feed almost all type of waste 2 An exothermic process and does not lead to any perceptible rise in the vermi reactor temperature 3 Technology promotes the growth of other essential bacteria and actinomycetes 2 Economically viable technology	1 Optimum conditions required due to sensitivity of earths worms towards heat, pH and moisture 2 Composting is difficult for meat, dead organisms, etc.	

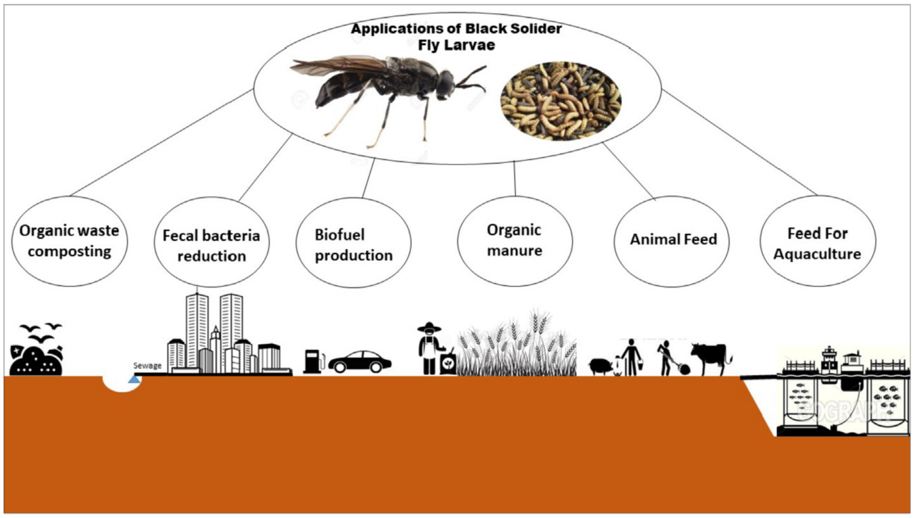

Fig. 3. Black Soldier Fly larvae as value chain developers.

findings reported. Nguyen et al. (2015) in his study also pointed out that the different dietary composition such as poultry feed, restraint waste, faeces, vegetables and flesh greatly affect the decomposition rates and development of larvae population.

Another interesting feature of BSF larvae is, they decompose the organic waste and subsequently generate nutrient residues which can be used as fertilizers. This in turn can have applications in agricultural activity just like compost or biogas production which can smartly contribute to the revenue of the country. A group of authors found that flies feeding to organic waste could facilitate the bioconversion of 30 tonnes of waste per day, accompanied by production of 33.3% residue that can be utilized as fertilizers and 7.7% of prepupal biomass which can be used for animal feeding. Green and Popa (2012) and some other groups of people have also reported that the larval manure possesses similar qualities as the fertilizers, wherein it can be utilized as composed for crop cultivations (Diener et al., 2011a; Choi et al., 2009; Green and Popa, 2012).

However, it was also observed that BSF larvae mediated treatment and valorisation has global warming potential to some extent but minimal in comparison to other conventional technologies (Salomone et al., 2017). Nevertheless, it has a major advantage in terms of minimal land requirements for food or energy production. As a supportive study, Perednia et al. (2017) also reported less amount of greenhouse gases emission (carbon dioxide and methane) in comparison to microbial decomposition, as they promote carbon sequestration in its own. The microbial decomposition process such as composting releases approx. 70% higher greenhouse gases (carbon based) and on the other hand these magnetic species captures and stores most of the released carbon in the body in form of nutrients such as proteins, lipids, chitin and fats possessing economical value as compared to the compost (Perednia et al., 2017).

6. Waste to valuable biomass

The BSF larvae are highly efficient in biomass conversion to human or animal feed and can be harvested easily without any dedicated facilities and of course are not pestiferous. Moreover, they do not accumulate myotoxins or pesticides in their body as conferred by many scientists therefore may be well recommended for animal and human consumption. However, the production methodology has some social stigma attached to it prohibiting the eating of organisms (Wang and Shelomi, 2017).

6.1. Fish meal or cattle meal production

BSF larvae are impeccable decomposers or converters of organic waste (or manure) to valuable biomass and can itself be used as dried supplemented diet for fish meal. They render nutrient benefits more efficiently than fish meal, therefore can be incorporated for products making, replacing the fish meal for animal feeding (Newton et al., 2005b). Cummins et al. (2017) in his study replaced traditional fish meal of marine fishes with about 25% of BSF larvae. Similarly, Magalhaes et al. (2017) concluded no difference in the growth patterns of European Seabass fed with BSF larvae wherein traditional meal was replaced with about 19% of BSF larvae. Supplementing the larvae with heterogeneous waste can even enhance the nutrient composition of larvae rather than homogenous feed source. In the same scenario, fly eggs subjected to mixed diet of fish offal and cow manure resulted in increased lipid secretions (43%) by the larvae as compared to single source of diet which included cow manure only. Studies are also available reporting the increased levels of omega-3 fatty acids increased from trace amounts to 3% (St-Hilaire, et al., 2007b). BSF larvae instead of being consumed as a whole can also be exploited for different nutrient compounds (fats, proteins, lipids and chitin) and sold separately enhancing the economic value of the species.

Another fascinating feature is that the chitin present in the cuticle of these flies (like the other insects) can be of commercial interest because of its high nitrogen composition (6.9%) as compared to synthetic cellulose (1.25%) (Diener et al., 2011b). Kumar (2000) in his study claimed that chitin renders its wide application in fields of medicine and biotechnological advancements. The average larval composition of BSF is shown in Fig. 4 which is based on the dry weight of insect (BSF) larvae meal (ESR International, 2008). Furthermore, several other studies have reported the diet specific nutrient composition (crude protein and fat) of larvae and is shown in Table 3. In addition, assessing the nutritional aspect of these flies, Cummins et al. (2017) and Caligiani et al. (2018) also reported carbohydrates (37%), chitin (9%) and lipids (15%) accumulation along with proteins (32%) in purchased BSF larvae. Earlier, Webster et al. (2016) also reported 10% lipid accumulation and 41% proteins in larvae fed with Menhaden fish substrate.

Recent diminishing global production of fishmeal have put the commercial market, poultry farmers and aqua culturists under pressure to develop new protein sources due to high price of available sources. The studies reported BSF larvae to be the efficient alternative source replacing fish meal, oil and conventional fat and protein sources to feed poultry animals and fishes (Hale, 1973; Newton et al., 1977; Bondari and Sheppard, 1981; Sheppard et al., 2007). Ferrarezi et al. (2016)

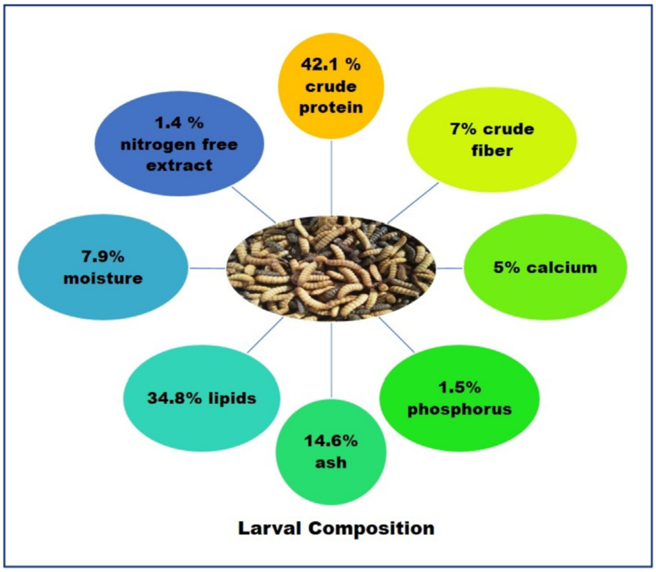

Fig. 4. The average BSF larvae composition (ESR International, 2008).

Table 3
Larvae nutritional composition (crude protein and fat) based on per dry weight of insect meal.

S. No.	Feeding source	BSF larvae composition (%)		References
		Fats	Crude Proteins	
1	Poultry manure	18.73	37.9	Arango Gutierrez et al., (2004)
2	Swine manure	28	43.2	Newton et al.
3	Poultry manure	34.8	42.1	(2005a, 2005b)
4	50/50 Fish offal: Cow manure	30.44	–	St-Hilaire et al., (2007a)
5	Animal manures	31–35	42–44	Yu et al. (2009)
6	Food manufacturing by-products	21–35	38–46	Oonincx et al. (2015)
7	Biogas digestate	21.8	42.2	Spranghers et al.
8	Vegetable waste	37.1	39.9	(2017)
9	Fresh fruit waste	41.7	37.8	Mutafela (2015)

stated that the utilization of larvae in feed production for fishes (Tilapia) is economically viable yet the economy of scale needed to be enhanced and further determined to generate enough larvae to act as a supplement as it can be an excellent cost-effective alternative for fish meal production in comparison to meal created from fishes. BSF larvae are used as protein feed for channel Catfish *(Ictalurus punctatus)* and blue Tilapia *(Oreochromis aureus)* by Bondari and Sheppard (1987) and other authors. In countries like India, where tropical environment prevails, the BSF species can be cultured easily and may be served for different applications such as in formulated diets, feed for cockerels, fishes like Tilapia and catfish (Hale, 1973; Sheppard and Newton, 2000; Sheppard et al., 2002). Kroeckel et al. (2012) and Makkar et al. (2014) used BSF larvae as the fish meal substitute for feeding turbot and channel fish as well. In addition, he also concluded that BSF larvae or prepupae can be used as alternative protein feed source for partial replacement of fish meal but are accompanied with low nutritive value. Other species such as rainbow trout, catfish, swine and chicks fed with larvae as a dietary component were also found to have good growth and development in various studies (Hale, 1973; St-Hilaire et al., 2007a). Similarly, Sealey et al. (2011) conducted sensory analysis in rainbow trout replacing fish meal with BSF larvae. The diets were supplemented with 25–50% fish meal or fish offal with larvae and there was no significant difference observed between the fish fed with BSF larvae and fish meal. Findings of Nyakeri et al. (2017) also revealed that these larvae can be a potential source of proteins for small scale production for poultry or fishes.

The greatest advantage of BSF larvae in feed production may be counted upon its lower environmental impact as compared to any other animal meat as it has lower feed to protein conversion. Oonincx et al. (2015) and van Huis (2013) has reported that BSF larvae have lower feed to protein ratio than swine, cattle or poultry animals. The larvae are considered a high-grade animal feed worldwide due to their high lipid and protein contents. Blending fish meal with this insect larvae renders high nutrition to the fishes. Renna et al. in his study showed that diet supplemented with 40% larvae showed no negative effect on physiology of rainbow trough. However, decrease in level of desirable unsaturated fats was observed (Renna et al., 2017). A similar study of diet supplementation with BSF larvae for juvenile Jian carp showed no difference on growth performance as compared to carp fed with soyabean, yet the lipid content was decreased with increased BSF larvae oil in diet (Li et al., 2016). The BSF larvae are also used for feeding aquatic invertebrates such as Shrimps replacing fish meal (Cummins et al., 2017).

This species is also being utilized for poultry feed replacement such as soyabean or maize feed and the reason is mainly the high potential of larvae to reduce the poultry waste. In the same perspective, authors have reported that the broiler quails and chicken (*Coturnix japonica* and *Gallus gallus domesticus*) fed with BSF larvae diet showed excellent yields and increased amino acid composition of meat but also accompanied with increased the concentrations of less required saturated fatty acids (Cullere et al., 2016, 2018). Thus, it can be concluded that BSF larvae can act as an impeccable source of poultry feed with additional benefit of rearing the species on same poultry waste and capable of recycling and valorising the waste. Likewise, the species can be reared on swine or dairy cow manure, however, the dairy cow manure is often mixed with other materials to avoid consequences related to high fibre content and low digestibility of the species particularly to fibres (Nguyen et al., 2015; Rehman et al., 2017a, 2017b). Thus, BSF larvae offers remarkable approach to manage vertebrate waste and valorise the waste to products for human use and animal feed.

However, in context with food safety and regulation, there are more stringent rules to avoid risks associated with insect consumption. Countries having no history of insect's consumption such as Europe are even stricter about regulations to mark the insects as novel food. UN's Food and Agriculture organisation has identified six species of insects including BSF as the safe species for human and animal consumption without any pestiferous impact (Wang and Shelomi, 2017). Similarly, in USA, FDA (Food and Drug Administration) along with AAFCO (Association of American Feed Control Officials) looks after the safety labelling, distribution, import and export of BSF larvae as feed grade materials for fishes and poultry (Association of American Feed Control Officials (AAFCO), 2018).

6.2. Black Soldier Fly for human consumption

Human consumption evidences of BSF larvae are very strenuous and troublesome. People consuming insects as food hardly bother to know what species they are eating or what is their scientific names as of course they are not entomologist. One example of Malaysian community is evident where local people consume about 60 species of insects and BSF being one of them. These species are collected from the fermented beverage and eaten raw. However, the culture of insect eating is diminishing gradually and grasshoppers, honey bee brood, and sago grub are more popular among them in comparison to BSF (Ramos-Elorduy, 1997; Chung et al., 2002; Durst et al., 2008). There are also Mexican findings reporting the human intentional eating of BSF larvae or prepupae (Mitsuhashi, 2017).

Consumer interest of larvae purchase or larvae oil from companies mostly lie in the animal feeding rather than human consumption.

However, based on limited studies or authors it was concluded that BSF larvae has fishy and pungent aroma and taste like earthy, chocolate or malt flavour and melting texture. Oil even has more pronounced malty flavour. Some others on the contrary reported less tasty which may be provisionally due to cooking methods (Nagy, 2017).

6.3. Biomass to energy production

Till date, the demand for energy and fuel has substantially increased and expected to increase more with increasing population and developments especially in low- or middle-income industries. According to source of EIA (2013), the worldwide energy demand is expected to increase up to 820 Quadrillion Btu (Q Btu) in 2040. Conventional fossil fossils being in debate for climate change and global warming related issues and has been restricted in use. In turn, the concept of biodiesel (a renewable fuel) is popularised worldwide from the last two-three decades. However, utilising edible crops for biodiesel production greatly compete with food demands and are highly unacceptable (Zheng et al., 2012). Consequently, the cost of production is another limiting factor which urges to develop other cost-effective alternatives for biodiesel production (Kumar et al., 2018). The high potential of BSF larvae to convert ligninocellulosic biomass to valuable products offers an imperative solution for biodiesel production from cheaper raw materials. Zheng et al. (2012) concluded that BSF larvae assisted with microbes fed with a mix substrate of rice straw and restaurant solid waste in the ratio of 30% and 70% produced BSF larval grease which was thereafter synthesised for biodiesel production. About 43.8 g of biodiesel was produced from 2000 larvae subjected to 1000 g of substrate. The similar findings were also observed in the studies of Li et al. (2011a, 2011b) and Zheng et al. (2011). Other studies of utilising BSF larvae for biodiesel or other fuel production is shown in Table 4.

7. BSF larvae composting and sustainability

Insect biotechnology is gaining huge attention nowadays and has been outstandingly successful. The BSF, H. *illucens* is one such success story of insect biotechnology. The species is robust, tolerant and have non pestiferous properties as concluded by many studies. The insect has high feed conversion ratio and high composting efficiency and not only restricted to that but also the bioactive substances in the larval extracts illustrates the future possibilities for successful insect mass culture which can be used for human and animal welfare (Muller et al., 2017). The species can be grown and harvested without any dedicated facilities or high operational skills. The biggest advantage over other insects is their ability to convert waste into food, generating value and closing nutrient loops as they reduce pollution and costs (Wang and Shelomi, 2017). Studies recommend BSF larvae composting is highly sustainable as composting produce affordable manure and animal feed in an environment friendly way as rearing of insects is more environment friendly as compared to livestock rearing with respect to water consumption/land requirement and greenhouse gas production (van Huis and Oonincx, 2017; Joly, 2018). A group of authors also called these flies “the dynamic creature” as they have numerous benefits and are skilled agents for composting applications and that too with minimal environmental intervention (Choudhury et al., 2018). They also exhibit the capability to be remoulded into a money-making entrepreneurial element paving new ways of income and employment opportunities in economically weak countries (Choudhury et al., 2018). Successful and established data pertaining to ability of BSF larvae to have economical valuable outputs is abundant in scientific domain. Based on various research outcomes, this technology can be regarded as the environmentally and financially sustainable system promoting the high resource recycling and reuse efficiency and minimal environmental impacts (Lalander et al., 2015).

Table 4
Utilization of BSF larvae for production of biodiesel and biogas.

Type of Waste Substrate	Type of fuel	Amount of fuel (gram)	References
Diary manure (1248 g)	Biodiesel	15.80	Li et al. (2011a)
Chicken manure (1000 g)	Biodiesel	91.40	Li et al. (2011b)
Cattle manure (1000 g)	Biodiesel	35.50	Li et al. (2011b)
Pig manure (1000 g)	Biodiesel	57.80	
Food waste (1 g)	Biogas	0.22	Lalander et al. (2018)
Faeces (1 g)	Biogas	0.13	

8. Constraints and research gaps

The major outcomes of the study revealed that the BSF larvae mediated treatment technology has lower impacts in terms of greenhouse gas emissions and land requirements but studies have also reported higher environmental impacts in terms of transport, electricity consumption during composting and feed production, and energy consumption for bioconversion processes. However, the environmental impacts of BSF larvae feed processing involve the GWP (Global Warming Potential) which is higher than other conventional processing methods but has a lower GWP when compared with microbes assisted composting. Minimal land requirements for insect rearing or production is also the most significant benefit over others. *The future challenges in research may broadly include:*

1. The technically sound data generation on biological treatment efficiency of BSF larvae mitigating the flaws encountered during the process and the optimum environmental conditions to have enhanced performance of larvae in treatment of waste which may further increase the financial output.
2. In addition, economy of scale is to be examined meticulously to produce sufficient quantity of larvae to replace or supplement animal feed need.
3. Sterilization of eggs and keeping the environment uncontaminated is also the major challenge.
4. Technical feasibility and scaling up of BSF larvae system are not extensively studied in developing countries which remains unclear as to what extent the result from laboratory studies can be employed to field studies which is a prerequisite for next generation industrial and commercial application of this technology.
5. The challenges associated with promoting global entomophagy that requires improved methods of rearing, alleviating the risks of insects' consumption, etc. is also of major concern (Shelomi, 2016).

Overcoming the challenges, will strengthen or reinforce the resilience of BSF larvae treatment system and waste management. Integrating the BSF units to treatment plants will further strengthen the waste management plans and policies. But, this in turn, demands iterative research to identify loop holes and strengthen the knowledge resource networks to have proper technical refinement of layout and operational facility to optimise profitability.

9. Conclusion

9.1. Summary

Recently, insect rearing as a protein or animal feed alternatives is becoming extraordinarily successful. Based on the findings, BSF *(Hermetia illucens)* has remarkable potential in field of solid waste management associated with livestock feed generation and biofuel production. The authors have assessed the treatment efficiency of BSF larvae for organic waste by 50–78% of its original amount. A comparison between different biological methods was made and supremacy of BSF larvae technology was also established in this study. The BSF larvae composting was found to be more sustainable and cost effective considering less sophisticated and easy operation, low technical skill requirements, land requirements/low ecological footprints and more adapted economical potential. The present study has also defined the correlation between the larval composting efficiency and environmental factors (temperature, relative humidity, moisture, pH and light) and it was found that species was able to survive in hostile environmental conditions such as drought, oxygen deficiency, and food shortage but to a certain extent. Moreover, an optimum range of temperature (27–37 °C), moisture (40–60%) and 60% relative humidity were prerequisites influencing the mating and oviposition of the species. The larval development was also affected by the pH and it was also found that the pH above 6 is optimum but at the same time larvae were able to regulate the pH up to 9. Sunlight in natural environment plays important role in mating events of the species and the findings revealed that 85% mating events occur in sunlight at an intensity of 110 μmol m^2s^{-1} and this is the reason why mating is greatly restricted in winter season. The findings also concluded mimicking the artificial conditions of light using Quartz iodide lamps or generating the same intensity as that of sunlight was very helpful in achieving the similar results. Assessing the global distribution of the fly, the species have cosmopolitan dispersal in tropical and warm temperate regions. The feeding patterns were also evaluated in this study and it was concluded that the feeding occurs only at the larval stage and adults rely on fat reserves. They are the obligate consumers of decaying organic matter such as food waste, animal material and faecal material.

Apart from this, the authors of this study have also evaluated the success story of BSF larvae technology as a tool to enhance biomass energy efficiency, as alternative animal feed or feed for human consumption. Based on the conclusion drawn from the above studies, the larvae are the excellent source of nutrient supplements as they have high nutritive assay accounting for 32–42% proteins, 30–37% carbohydrates, 15–43% lipid secretions, and chitin. Acknowledging the nutritive assay, the larvae can be regarded as the potential alternative for fish feed or livestock feeding. Furthermore, the bioactive and antibacterial properties (not covered in this paper) of Black Soldier flies can also be explored projecting new areas of research.

9.2. Study limitations

This review paper is mainly focused on the importance of BSF larvae in organic waste composting and management, their efficiency in organic biomass conversion. Besides that, animal feed production and bioenergy synthesis as an end product of BSF larvae applications to waste is also covered but the waste management is the major concern from the authors point of view and interest. Moreover, the literature also entails the life cycle study of insect and its survival aspects but does not include the detailed methodology adopted by different authors for waste composting. In addition, the literature also does not focus on the comparative analysis of relevant performance of BSF larvae in bioconversion of waste for different climatic zones, for instance, developing or developed countries; tropical or temperate climate conditions.

9.3. Recommendations and future directions

Sustainable and environmentally sound management options are the need of hour for any kind of management, whether solid waste or air pollution. Considering the solid waste management, plenty of biological approaches are in practice to achieve the high waste-to-biomass conversion ratio and creating value added chains by synthesizing biomass-based products. In this context, BSF larvae composting outshines the present conventional.

Methods by a substantial margin (Choudhury et al., 2018). Although, the ultimate success significantly depends on the social and political factors such as funds availability, technical feasibility, segregation inputs and waste generation rate (Wilson et al., 2015). The technology will help to overcome the problems of resource scarcity in terms of food and energy in developing countries. A way ahead, possible collaborations between private non-profit organizations and other small entrepreneurs may assure commercial success and policy related implementations in countries. Employing these fanatical feeders in frugal application towards the societal betterment can develop a new scope to explore feasible business opportunities and cope up with unemployment issues. But large-scale production and utilization of insect-based products still show considerable challenges in terms of psychological barrier of the society. Zhang et al. (2010) concluded that the technology transfer and market development may facilitate the implementation aspects of this technology. Yet, many uncertainties and data gaps still remain which demands the holistic approach for technology up gradation. Further research is needed to deal with challenges and to validate the assumptions.

Conflict of interests

The authors declare no conflict of interest.

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